

Figure 1: Constraint Costs -- R25b (baseline) vs R28a

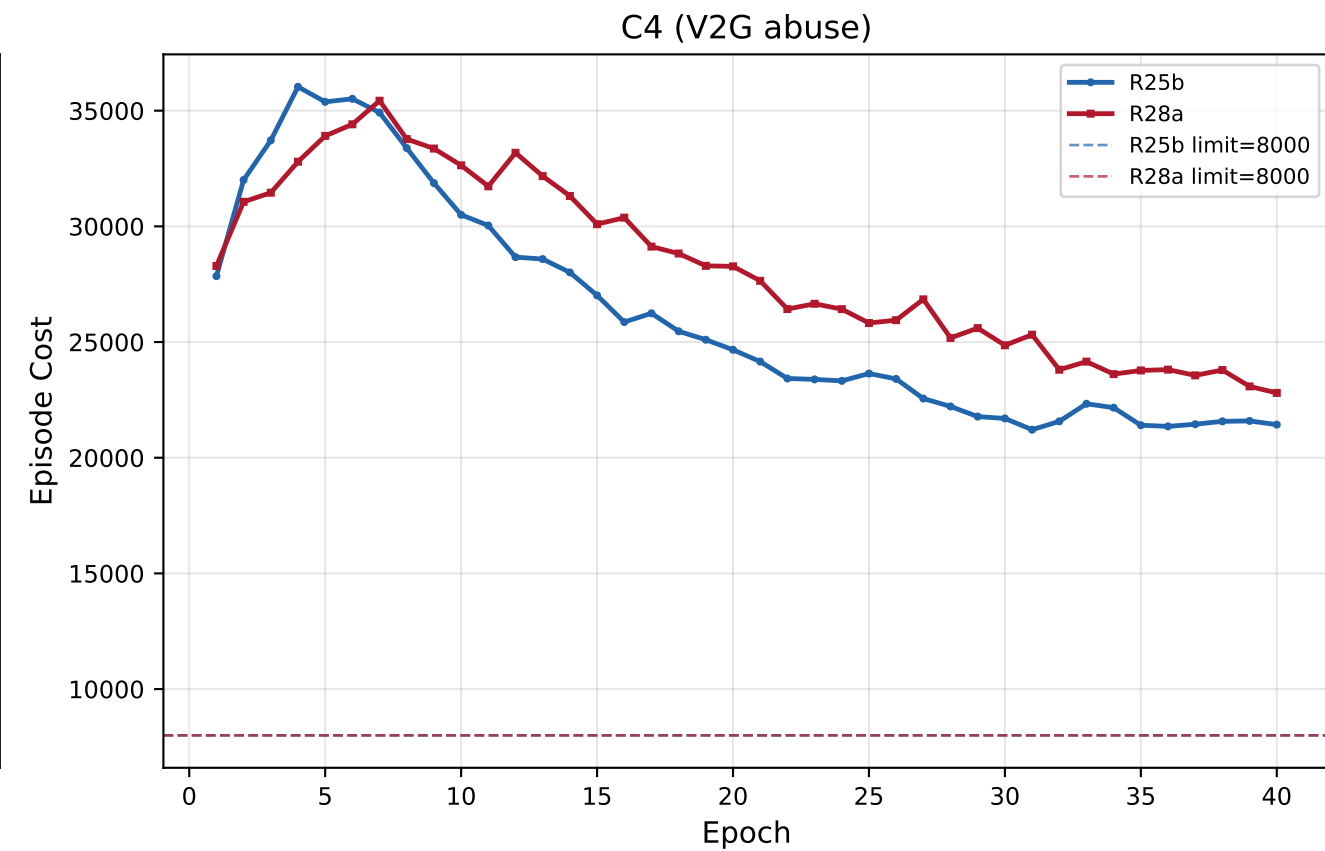
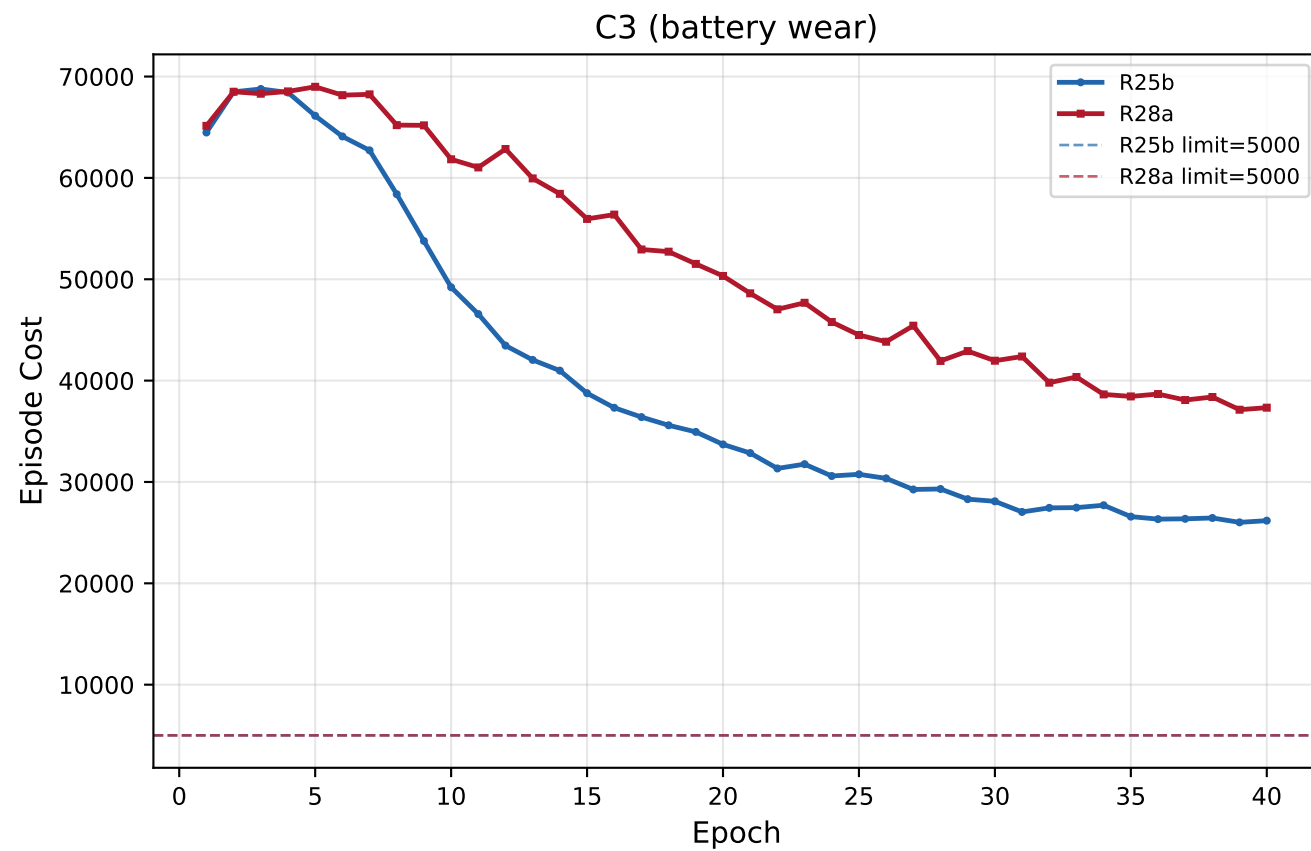
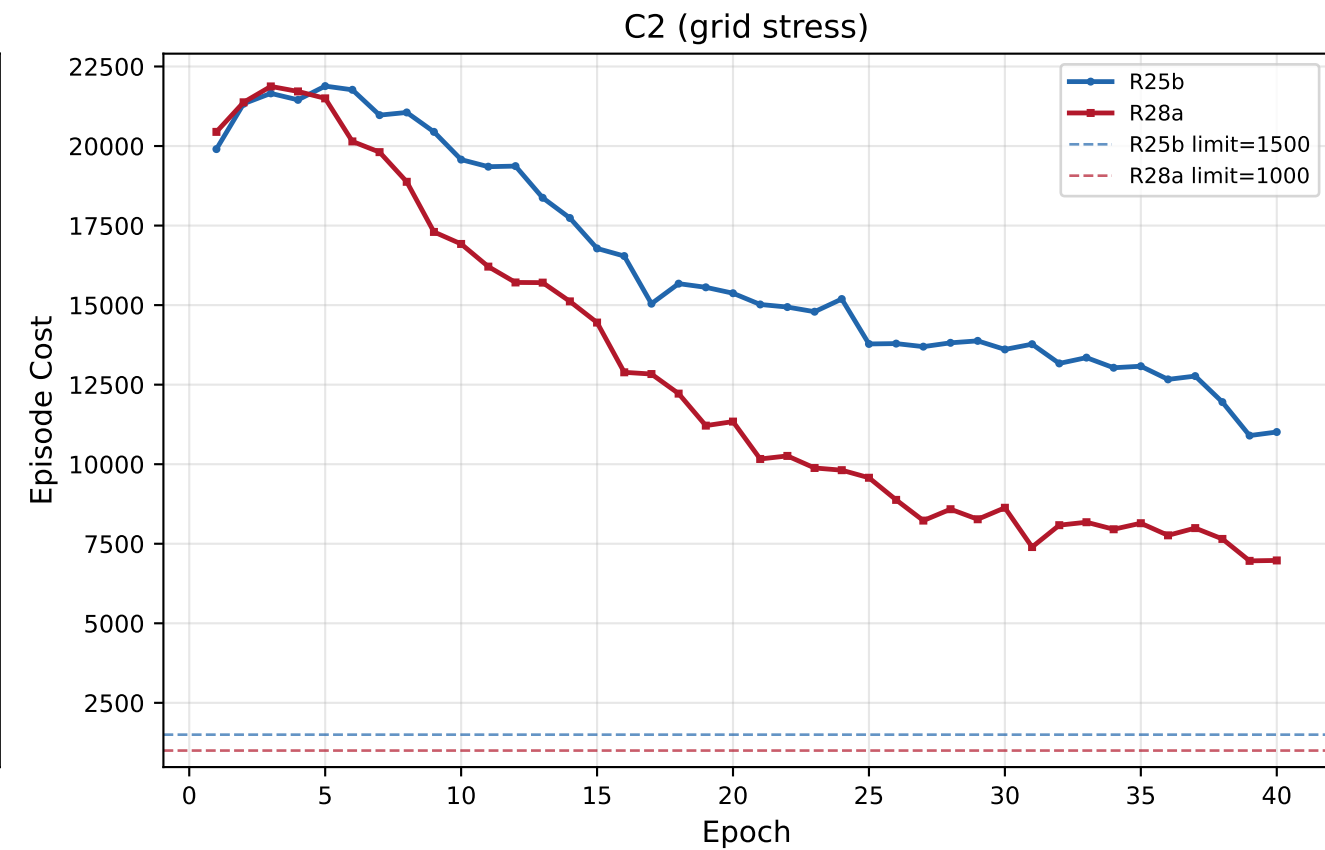
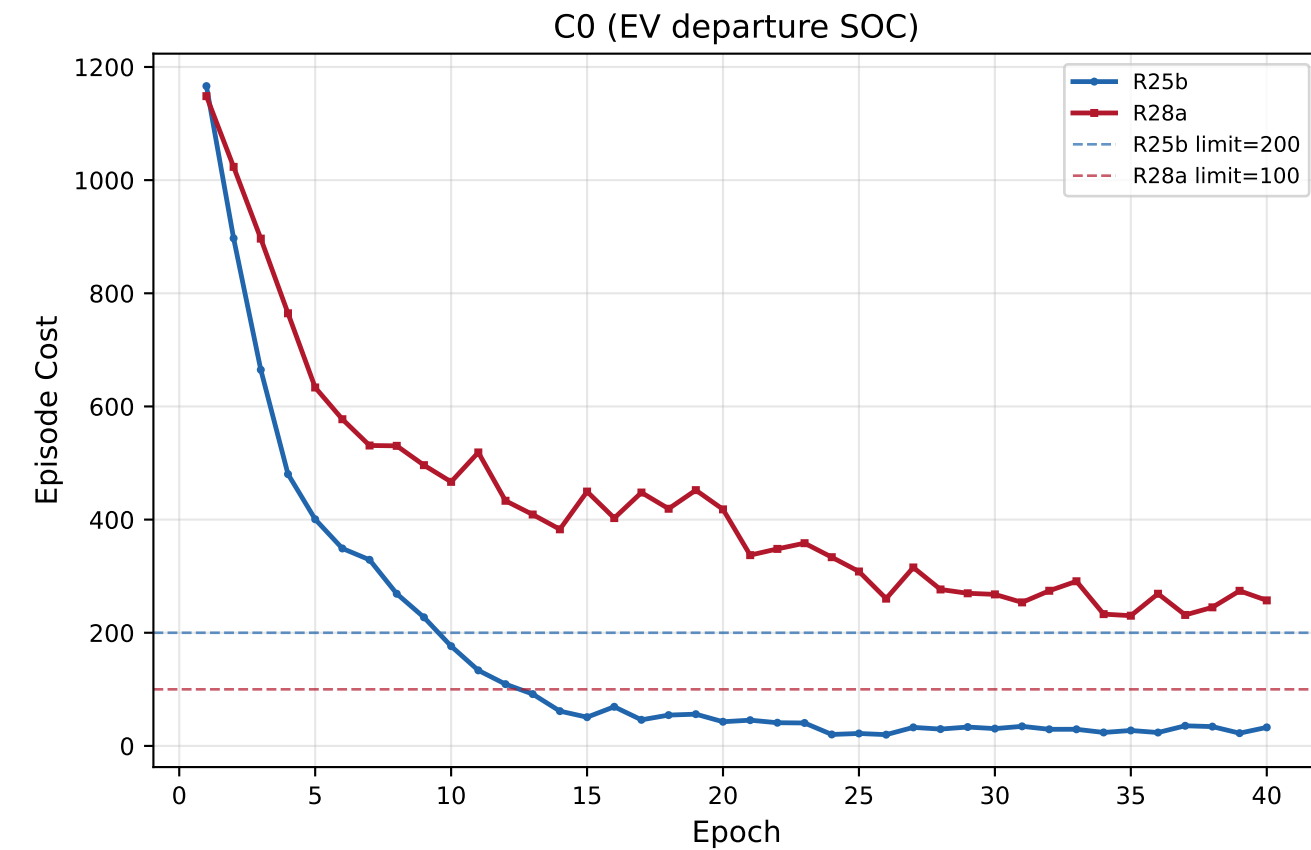


Figure 2: Lagrange Multipliers -- R25b vs R28a

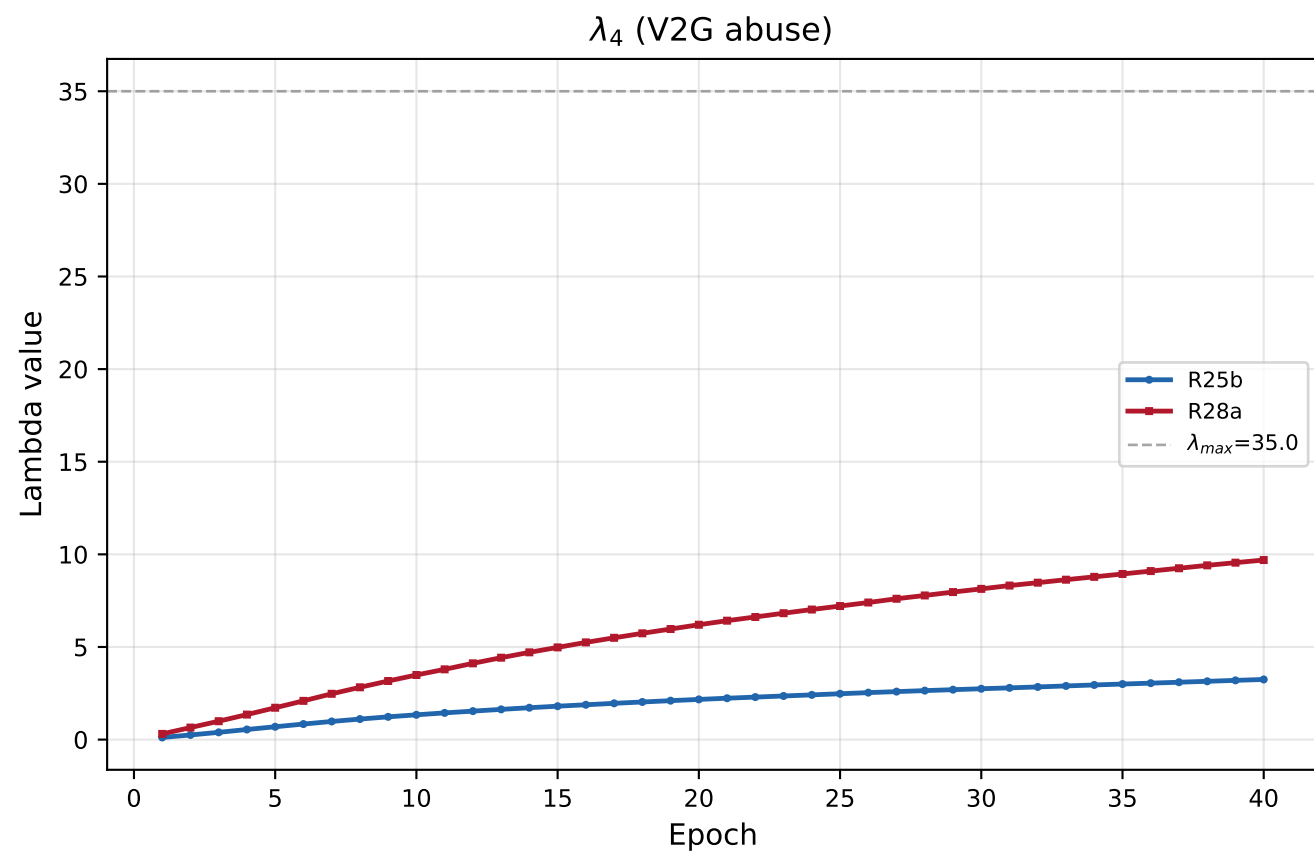
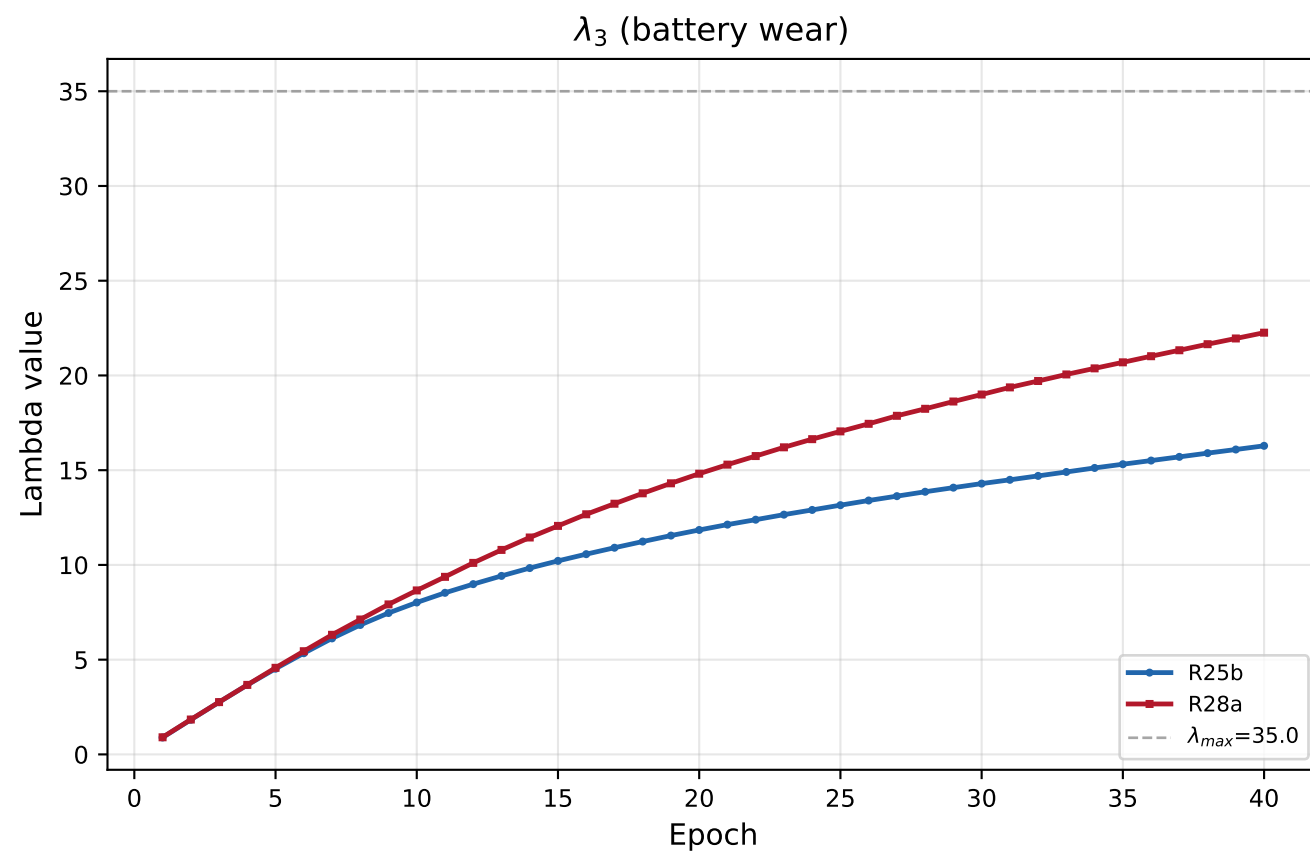
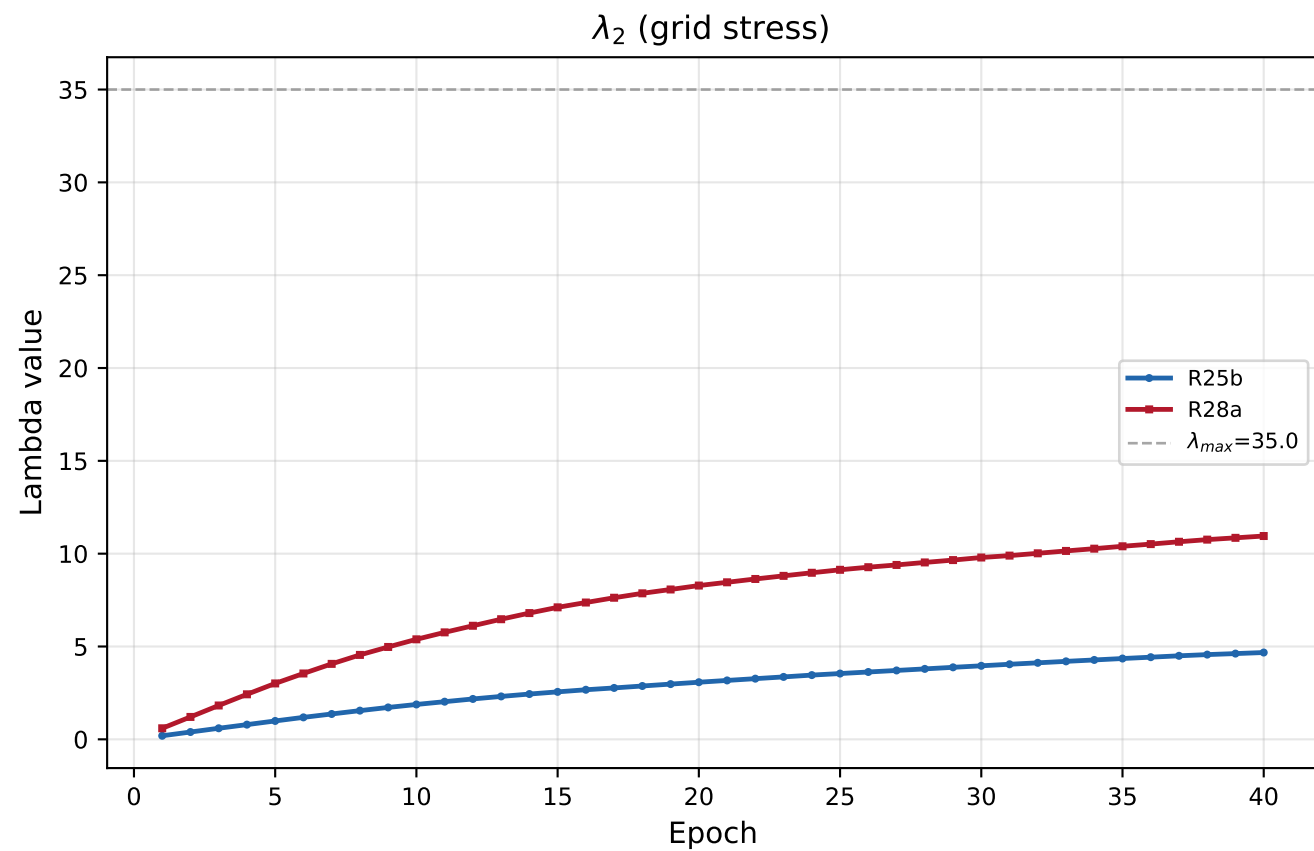
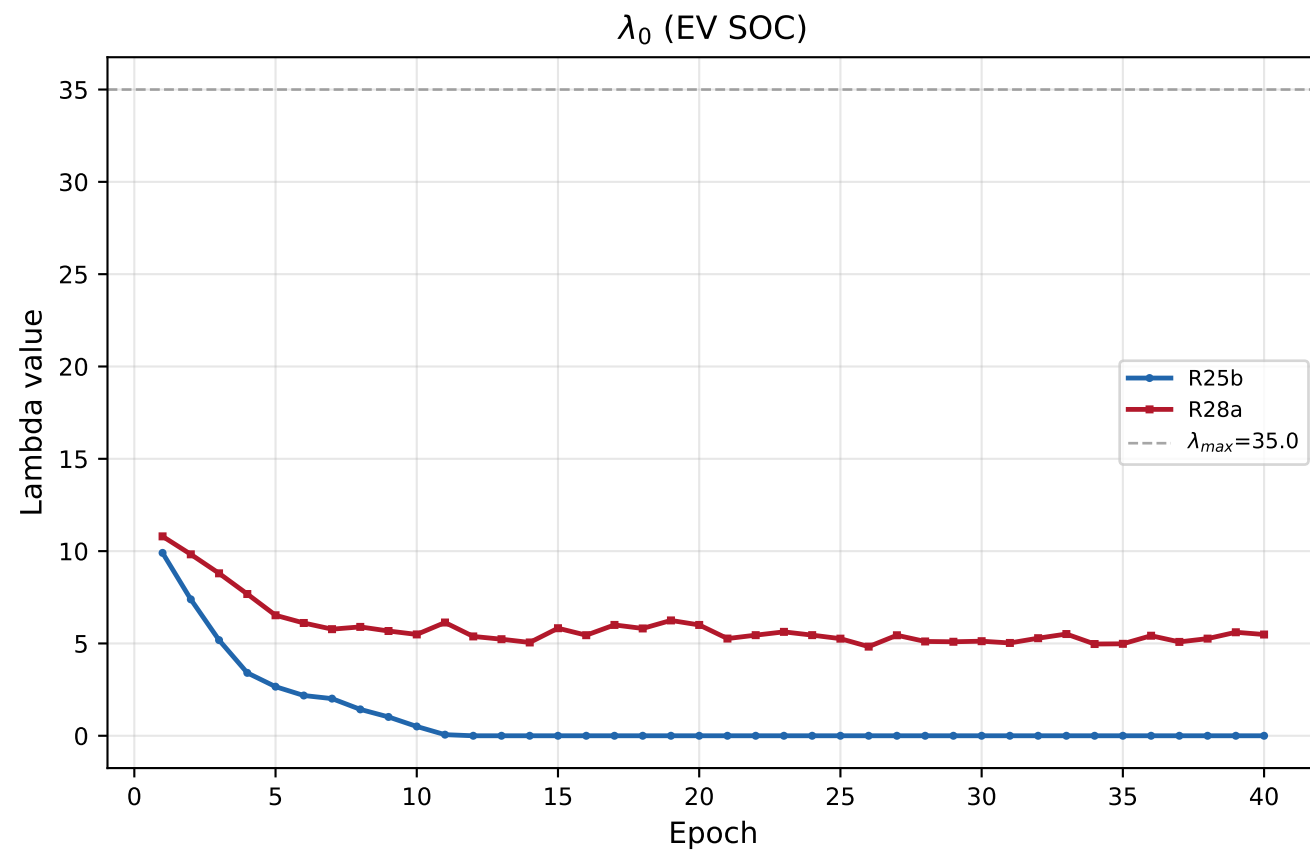


Figure 3: Per-Reward-Term Comparison

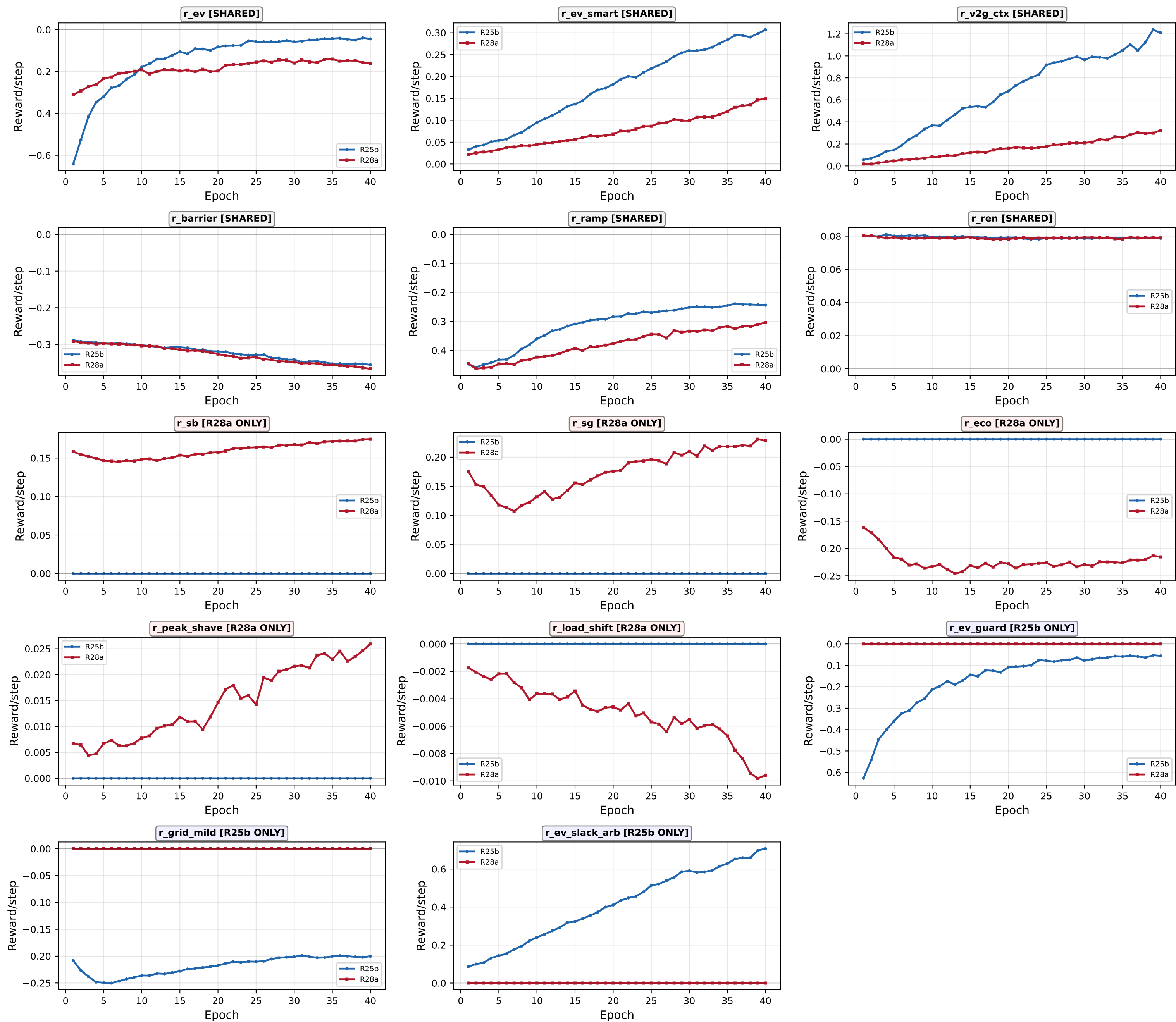


Figure 4: Total EV Reward Signal (sum of all EV-related terms)

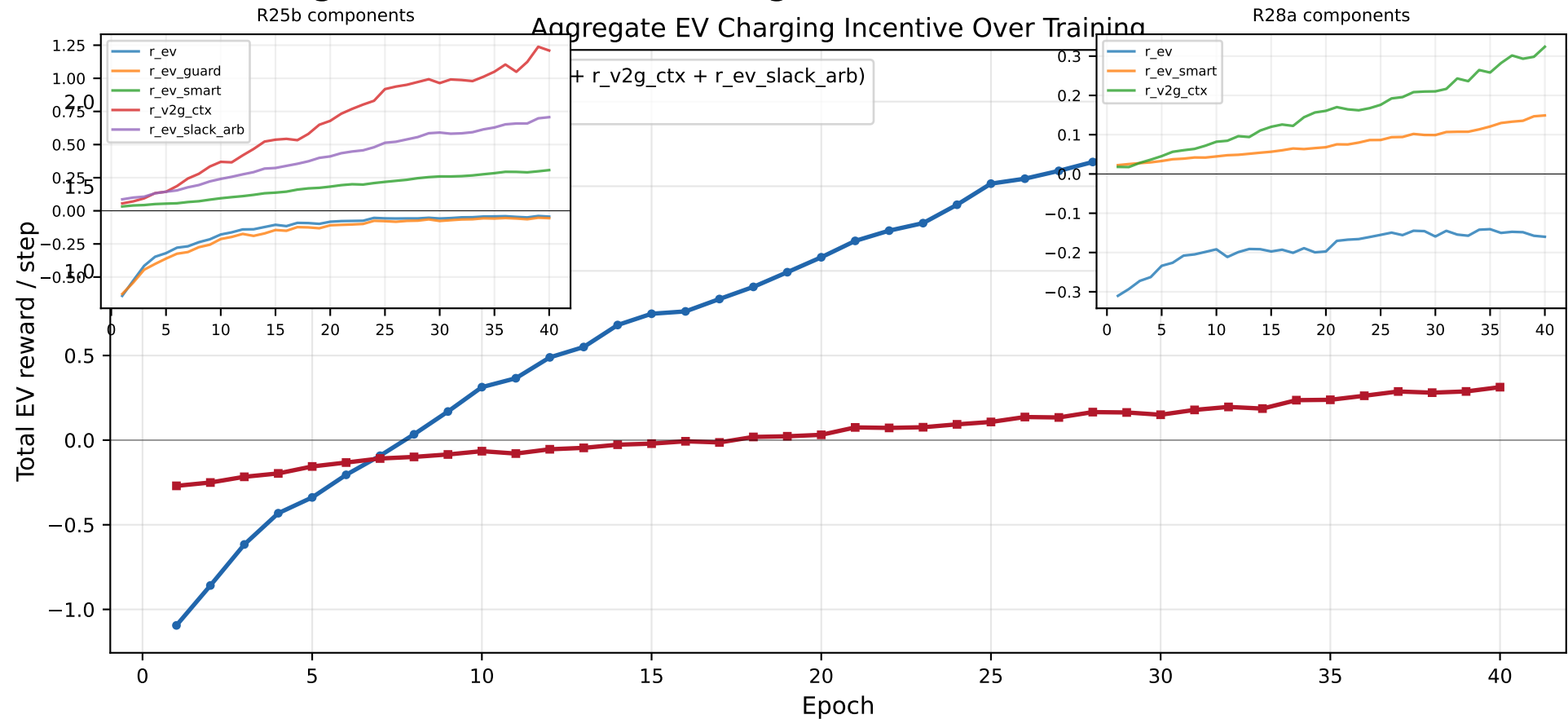


Figure 5: Training Health Indicators

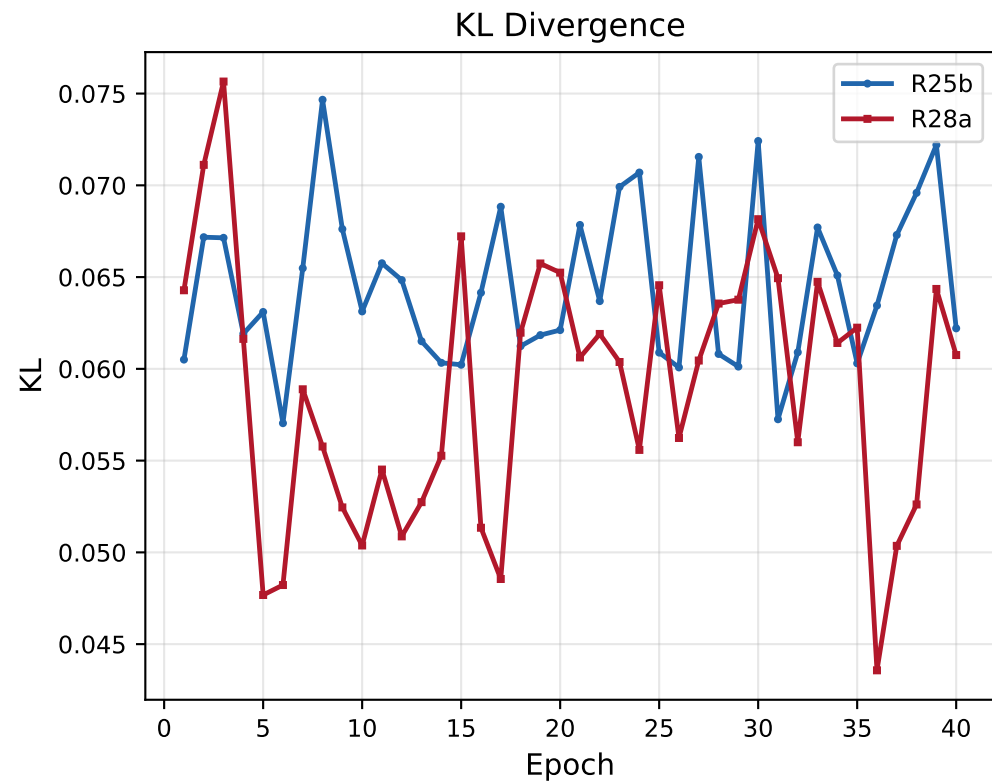
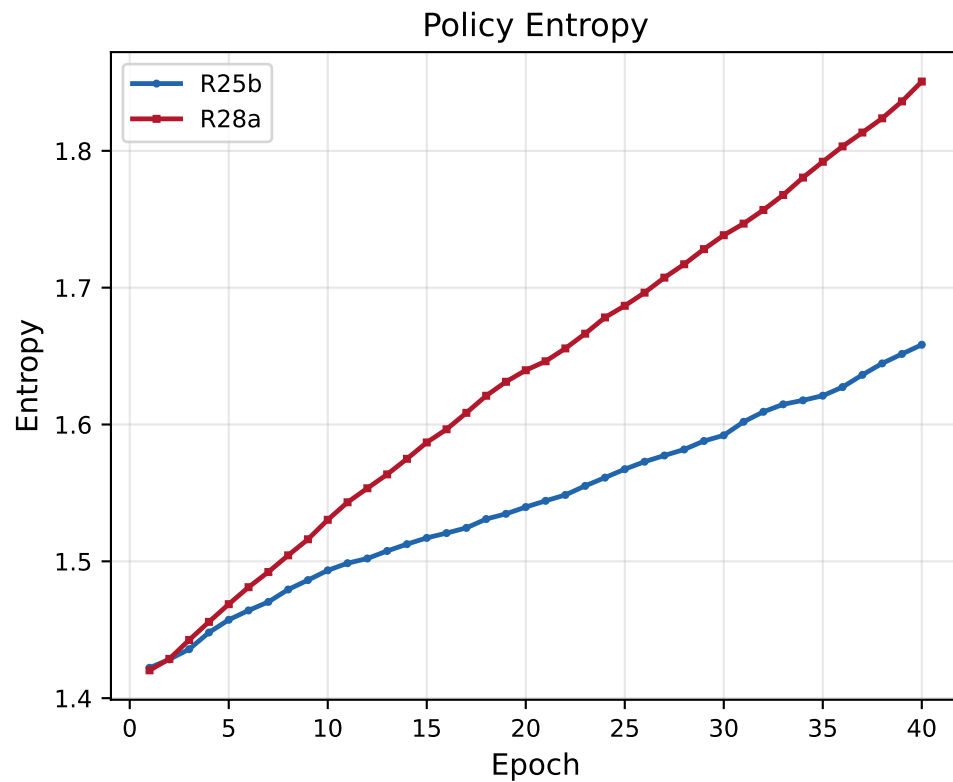
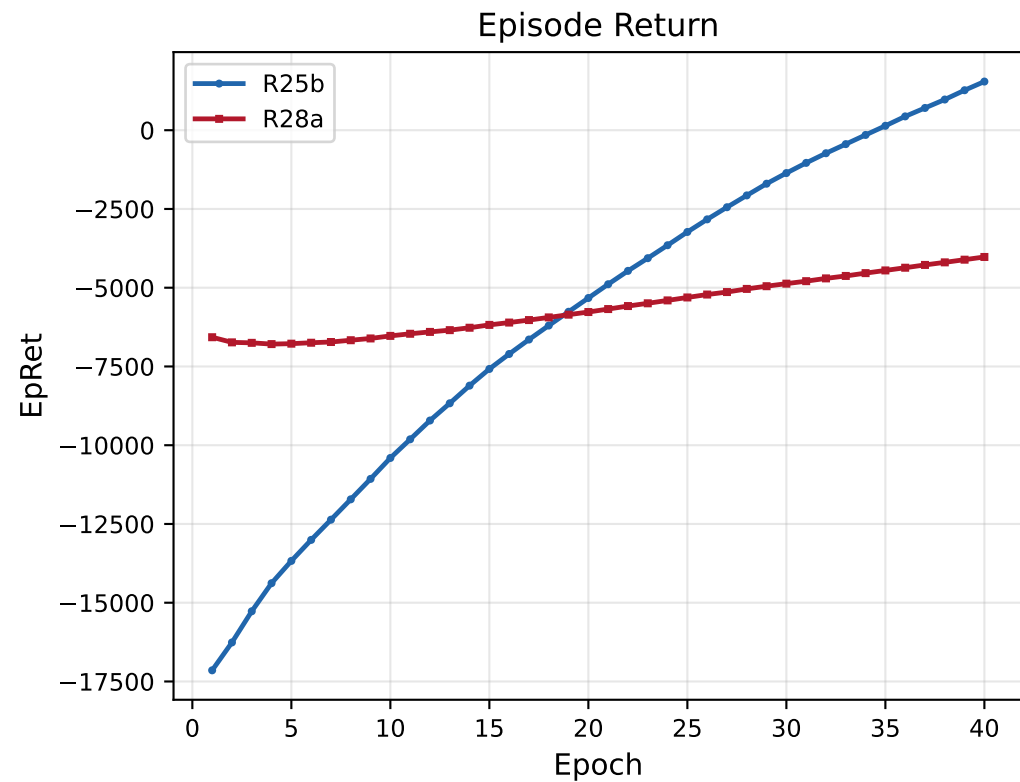


Figure 6: Normalized Violation Ratio (EpCost / Limit) -- lower is better

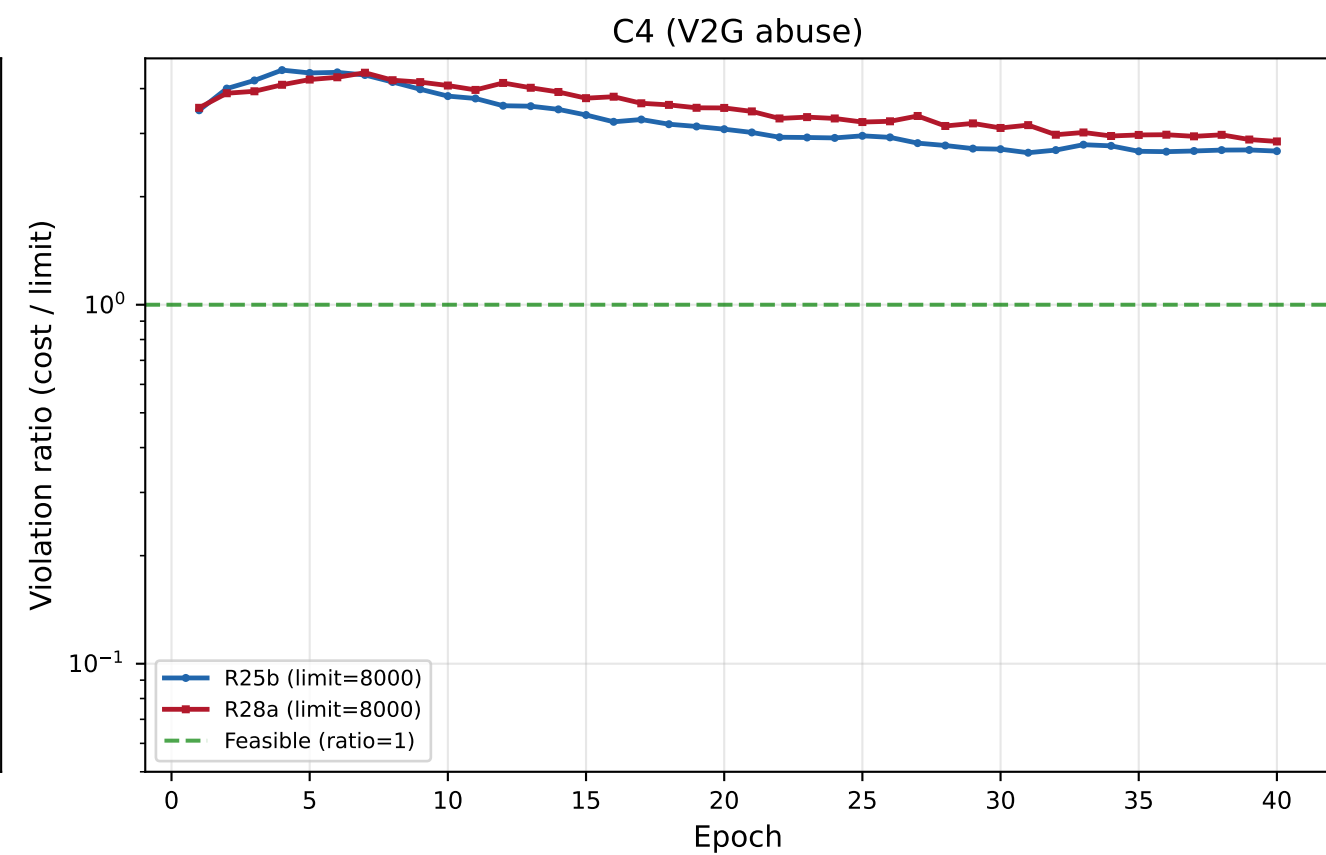
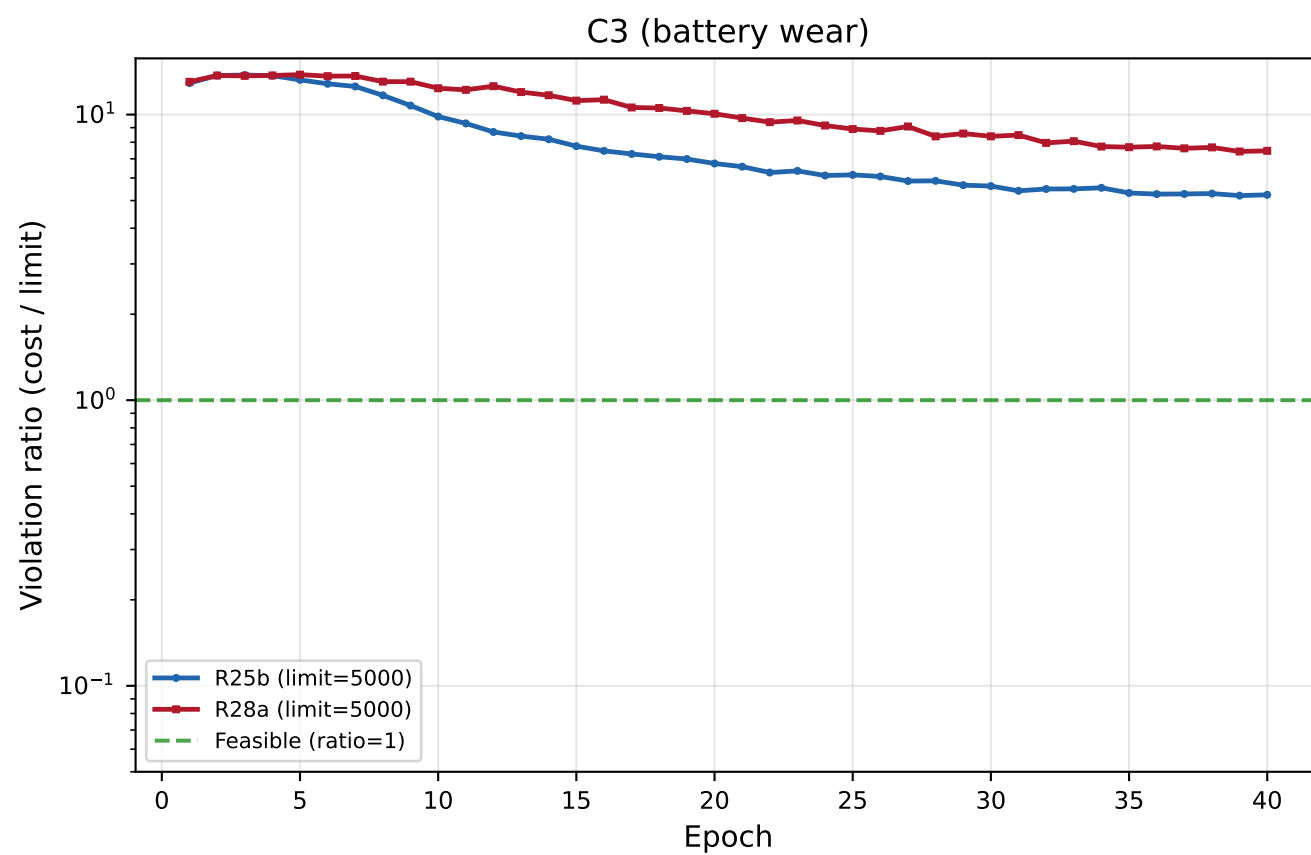
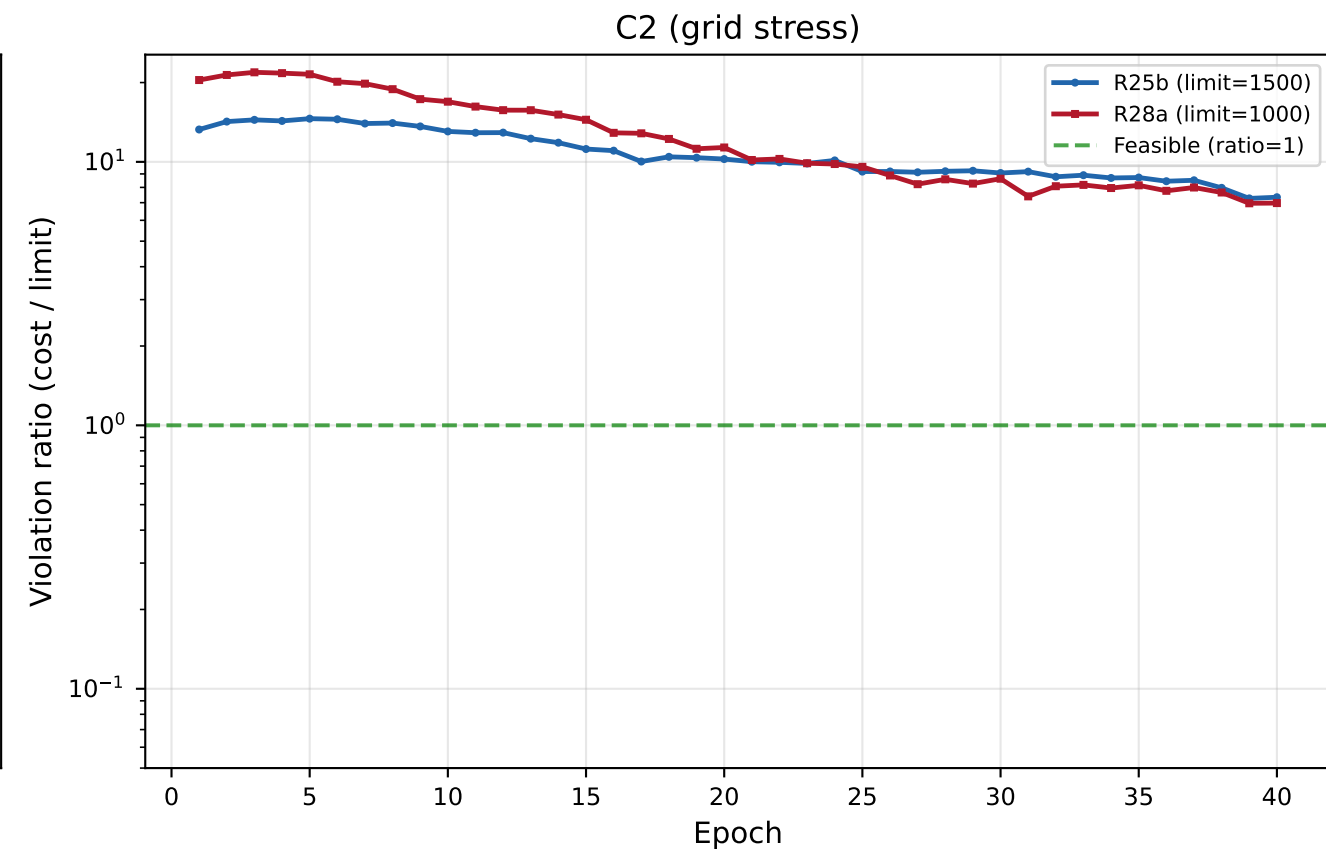
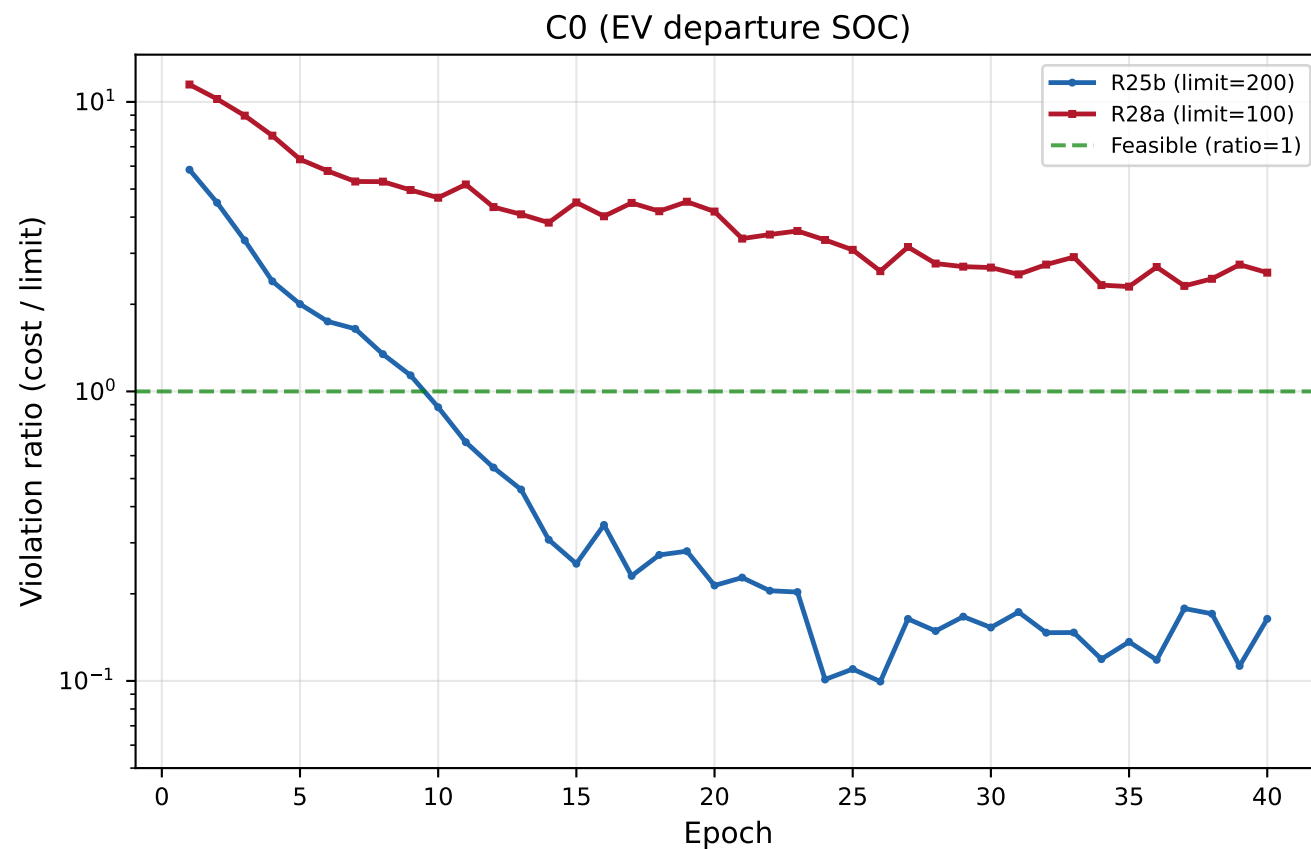


Figure 7: Summary Comparison Table

Reward Term Health Summary

Reward Term	Category	R25b Mean	R25b Status	R28a Mean	R28a Status
r_barrier	SHARED	-0.3226	ACTIVE	-0.3276	ACTIVE
r_eco	NEW	+0.0000	N/A	-0.2234	ACTIVE
r_ev	SHARED	-0.1415	ACTIVE	-0.1862	ACTIVE
r_ev_guard	REMOVED	-0.1677	ACTIVE	+0.0000	N/A
r_ev_slack_arb	REMOVED	+0.4102	ACTIVE	+0.0000	N/A
r_ev_smart	SHARED	+0.1775	ACTIVE	+0.0766	ACTIVE
r_grid_mild	REMOVED	-0.2184	ACTIVE	+0.0000	N/A
r_load_shift	NEW	+0.0000	N/A	-0.0050	DEAD
r_peak_shave	NEW	+0.0000	N/A	+0.0147	weak
r_ramp	SHARED	-0.3109	ACTIVE	-0.3783	ACTIVE
r_ren	SHARED	+0.0793	ACTIVE	+0.0789	ACTIVE
r_sb	NEW	+0.0000	N/A	+0.1593	ACTIVE
r_sg	NEW	+0.0000	N/A	+0.1748	ACTIVE
r_v2g_ctx	SHARED	+0.6706	ACTIVE	+0.1568	ACTIVE

Constraint Convergence Summary

Constraint	R25b Init	R25b Final	R25b Ratio	R25b 1st Feasible	R28a Init	R28a Final	R28a Ratio	R28a 1st Feasible
C0	1166	33	0.2x	10	1148	257	2.6x	never
C2	19904	11012	7.3x	never	20445	6975	7.0x	never
C3	64482	26184	5.2x	never	65135	37330	7.5x	never
C4	27849	21431	2.7x	never	28289	22801	2.9x	never